

Problem

A circuit is described by

$$4 \frac{dv}{dt} + v = 10$$

(a) What is the time constant of the circuit?

(b) What is $v(\infty)$, the final value of v ?

(c) If $v(0) = 2$, find $v(t)$ for $t \geq 0$.

Step-by-step solution

Step 1 of 4

$$\text{Let } v = v_h + v_p, \quad v_p = 10$$

$$v_h + \frac{1}{4}v_h = 0 \rightarrow v_h = A e^{-\frac{t}{4}}$$

$$v(0) = 2 = 10 + A \rightarrow A = -8$$

$$v = 10 - 8 e^{-0.25 t}$$

Step 2 of 4

$$(A) \tau = 4S$$

Step 3 of 4

$$(B) v(\infty) = 10V$$

Step 4 of 4

$$(C) v = 10 - 8 e^{-0.25 t}$$